

Marjan Mohammadi

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 [GitHub](#) | [LinkedIn](#)

Profile

Data Engineer and interdisciplinary researcher with a strong foundation in **software engineering, data pipelines, and machine learning for large-scale data**. Completing a Master's at the University of Toronto (Aug 2025), where I built scalable data workflows integrating AI and high-performance computing. Experienced in **ETL/ELT systems, SQL-based data handling, and cloud/cluster environments**. Passionate about building robust, reproducible, and high-throughput data infrastructure to accelerate scientific discovery.

Technical Skills

- **Programming & Data:** Python, Java, Bash, SQL, Git
- **ETL & Pipelines:** Automated data workflows on SLURM/HPC; exploring Airflow for orchestration concepts
- **Databases & Warehousing:** SQL (Oracle PL/SQL, academic datasets), structured data processing, exploratory analysis
- **Cloud & Containers:** Linux servers (HPC/Compute Canada), basic Docker use, AWS concepts exploration
- **ML/DL Frameworks:** PyTorch, scikit-learn, TensorFlow, Transformers
- **Visualization:** matplotlib, seaborn, BioRender, t-SNE, UMAP
- **Bioinformatics Tools:** JackHMMER, BLAST, MMseqs2, AlphaFold, DESeq2
- **Testing Frameworks:** Unit Test, Mockito, test-driven development practices from software engineering roles

Education

- **M.A.Sc. in Chemical Engineering (Machine Learning in Biology)**
University of Toronto, Canada | Sep 2022 – Present (Expected Aug 2025)
- **B.Sc. in Software Engineering**
Kharazmi University, Tehran, Iran | 2010 – 2014

Relevant Research & Projects

Data Engineering for Protein Design – Master's Thesis

University of Toronto

- Designed and optimized data preprocessing and embedding pipelines for large biological sequence datasets.
- Built workflows to integrate **ETL-like processes**: raw sequence input → cleaning → model-ready formats.
- Applied **SQL and Python** to manage structured biological data at scale.
- Visualized embeddings with **t-SNE and UMAP** to evaluate data quality and clustering.
- Managed compute-intensive workloads using **HPC clusters** (SLURM) and explored **Dockerized workflows** for reproducibility.

Sequence-to-Function Prediction Pipelines

- Developed predictive models to map protein sequences to functional properties.
- Implemented automated data transformations, normalization, and storage for high-throughput prediction experiments.

Data Integration & Automation in NGO Health Projects – Cordaid & INTERSOS, Kabul (2019)

- Built and maintained data collection pipelines using ODK and CommCare.
- Generated real-time health data reports with strong focus on accuracy and reproducibility.

Professional Experience

Enterprise Solutions Manager – Roshan Telecom Company | Kabul, Afghanistan | Nov 2019 – Sep 2021

- Led enterprise-scale IT projects; managed data-driven solutions integrating APIs and databases.
- Improved pipeline efficiency by applying **statistical and systems-level data analysis**.

Java Software Developer – Kavosh Core Banking Company | Tehran, Iran | 2016 – 2019

- Designed and maintained backend software using **Java, Oracle PL/SQL**, and **modular APIs**.
- Developed secure and scalable **data integration workflows** (Spring, Hibernate, Tomcat).

Java Developer – Tosan Banking Solutions | Tehran, Iran | 2015 – 2016

- Designed and tested backend systems for banking software using **Java frameworks**.
- Applied **testing frameworks** such as Unit Test and Mockito to ensure reliability and quality in software products.

Honors & Awards

- **DVF Award Winner (2022)** – Named one of 10 inspirational Afghan women.
- **First Prize – Code4Peace Hackathon (UN Women)**
- **First Prize – Startup Weekend (supported by Google)**